Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 3: Fibre to Fabric

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	3 — Fibre to Fabric
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors weaponise the most common Class-7 confusions — reversing process order, swapping which step does what, mixing up the silk-moth life-cycle stages, confusing fibre chemistry (keratin vs fibroin vs cellulose), and attaching the wrong reason to a correct-sounding fact. Do not write on this paper — mark answers on your answer sheet.

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1. A teacher holds up two unlabelled threads. Thread X dissolves and chars with a smell of burnt hair when flamed; Thread Y leaves a soft grey ash and smells like burning paper. Which conclusion is best supported?

(A) Thread X is a protein (animal) fibre such as wool or silk, while Thread Y is a cellulose (plant) fibre such as cotton
(B) Thread X is cotton because cotton smells of burnt hair
(C) Both threads are animal fibres because both burn
(D) Thread Y is silk because silk leaves a papery ash

2. Although wool and silk are both animal fibres, they reach us by completely different biological routes. Which contrast is correct?

(A) wool is a fluid the sheep secretes that hardens, while silk is the silkworm's body hair
(B) both are spun by insects that feed on mulberry leaves
(C) wool is hair that grows as a sheep's body covering and is sheared off, while silk is a fluid the silkworm secretes that hardens into a long filament
(D) wool is taken from a plant and silk from an animal

3. Two woollen blankets of equal mass are compared. Blanket P is loosely woven from highly crimped fibres; Blanket Q is tightly compressed felt. On a cold night, blanket P keeps a person warmer. The best explanation is that:

(A) blanket P is heavier, and heavier blankets are always warmer

(B) blanket Q has lost its protein and become cotton

(C) blanket P traps more still air among its crimped, loose fibres, and trapped air is a poor conductor of heat

(D) blanket P generates its own heat from the wool

4. A farmer in a hot, humid lowland and a herder in a cold Himalayan highland each want to rear sheep for the BEST wool. Based on NCERT, which choice is most sensible?

(A) The lowland farmer is better placed, because heat makes sheep grow more wool to stay cool

(B) The highland herder is better placed, because cold-climate sheep grow thicker fleece and India's wool sheep are reared in hilly regions like Jammu & Kashmir and Himachal Pradesh

(C) Both are equally good, since climate has no effect on fleece

(D) Neither can rear sheep, since wool comes only from goats

5. During wool processing, four steps are listed jumbled: dyeing, scouring, sorting, picking burrs. Which arrangement places them in the correct order as they actually occur?

(A) dyeing → scouring → sorting → picking burrs

(B) sorting → scouring → dyeing → picking burrs

(C) scouring → sorting → picking burrs → dyeing

(D) picking burrs → dyeing → scouring → sorting

6. Selective breeding of sheep over generations aims mainly to:

(A) develop flocks whose fleece is finer, softer and more uniform by breeding from animals with the best wool

(B) make sheep that produce silk instead of wool

(C) remove the fleece so sheep no longer need shearing

(D) turn the wool protein into cellulose for stronger cloth

7. The life cycle of the silk moth is given with one stage blanked: egg → _____ → pupa → adult moth. The missing stage, and what happens during it, is:

(A) larva (caterpillar/silkworm) — it feeds on mulberry leaves and grows

(B) cocoon — it lays the eggs for the next generation

(C) adult moth — it spins the silk thread

(D) pupa — it hatches directly from the egg before the larva

8. A student claims, 'The cocoon is just a single short hair the silkworm sheds.' The most accurate correction is:

(A) the cocoon is many short threads glued randomly, like wool felt

(B) the cocoon is one continuous silk filament, hundreds of metres long, that the larva spins and winds around itself

(C) the cocoon is the moth's outer fleece, sheared like sheep's wool

(D) the cocoon is a hollow egg shell left after the moth hatches

9. In Indian sheep-rearing regions, fleece is normally removed in spring/early summer rather than at the start of winter. Choosing the WRONG reason, which statement is NOT a valid justification?

(A) Shearing in early summer means the new fleece can regrow before the next winter's cold

(B) Removing the fleece in cold winter would leave the sheep exposed to the cold

(C) Shears only work in warm weather, so winter shearing is mechanically impossible

(D) The sheep do not suffer from heat loss when sheared in the warmer months

10. A care label warns: 'Wash in cool water, do not wring.' A child instead washes the wool sweater in hot water and scrubs hard. Predict the result and the reason:

(A) the sweater stretches larger, because hot water relaxes the fibres

(B) the sweater turns into cotton, because heat changes the protein into cellulose

(C) the sweater shrinks and mats, because heat and agitation make the scaly wool fibres lock tightly together

(D) nothing happens, because wool is unaffected by hot water

11. A factory accidentally tries to pick burrs out of greasy, unwashed fleece (before scouring). The most likely outcome is that:

(A) the grease and dirt hide the burrs, so many burrs are missed and the wool is poorly cleaned

(B) the burrs come out more easily because grease lubricates them

(C) the order makes no difference, since each step is independent

(D) the wool gets dyed automatically while burrs are removed

12. Scouring and reeling both use water on the fibre, but for opposite-sounding purposes. Which pairing of step to purpose is correct?

(A) scouring dyes the wool; reeling boils the silk to colour it

(B) scouring unwinds the silk thread; reeling washes the wool fleece

(C) both steps only add the final twist that turns fibre into yarn

(D) scouring washes grease and dirt out of raw fleece; reeling uses hot water to soften the cocoon's gum so the silk thread can be unwound

13. Tasar, eri and muga are all examples of:

(A) synthetic fibres made from petroleum, like nylon and polyester

(B) non-mulberry silks produced in India from silkworms other than the common mulberry silkworm

(C) grades of sheep's wool sorted by fineness

(D) the three life-cycle stages of the silk moth

14. A student asks WHY the silkworm larva spins a cocoon at all. Biologically, the cocoon's purpose is to:

(A) store food for the adult moth to eat after it emerges

(B) attract a mate while the insect is still a larva

(C) trap mulberry leaves so the pupa can keep feeding

(D) form a protective silk case around the larva while it changes into a pupa

15. In commercial silk production, cocoons are exposed to steam or boiling water BEFORE the adult moth emerges. The chief reason is that:

(A) the heat makes the pupa grow into a larger moth that spins more silk

(B) if the moth bit its way out, it would cut the single long thread into many short, less useful pieces

(C) boiling is mainly to wash dust off the cocoon for hygiene

(D) the heat dyes the silk a permanent golden colour

16. Why are several cocoons' filaments usually combined during reeling instead of using one cocoon's thread alone?

(A) a single filament is too thin and weak, so combining several gives a usable, stronger silk thread

(B) a single filament is far too thick to pass through a needle

(C) each cocoon gives only a different colour, so several are mixed for white

(D) one cocoon's thread is made of wool and must be blended with silk

17. Of all the people in the wool industry, sorter's disease (anthrax) most threatens the wool sorter rather than the finished-sweater shopper, because:

(A) the disease only attacks people who wear, not handle, wool

(B) the sorter handles and inhales dust from raw, unwashed fleece that can carry Bacillus anthracis spores, while the shopper's sweater is already scoured and processed

(C) sweaters are made of cotton, so shoppers face no risk

(D) the dye used on finished wool actively cures the disease in the body

18. Sorter's disease spreads mainly by one route into the body. Reasoning from that route, why does a dust mask protect a sorter more than rubber gloves alone?

(A) the spores enter only through cuts in the skin, so gloves alone fully protect a sorter

(B) the spores are mostly inhaled in raw-fleece dust, so a mask blocks the main entry route while gloves alone leave breathing unguarded

(C) the disease is caught by drinking sheep's milk, so neither a mask nor gloves helps

(D) the spores enter through the eyes, so only goggles would help

19. Match each animal to the fibre it provides. Which single pairing is correct?

(A) sheep → silk; silk moth → wool (B) mulberry plant → silk; sheep → cotton

(C) yak → silk; camel → cotton

(D) Angora goat → wool (Pashmina); silk moth → silk

20. An ethical objection to traditional (mulberry) silk-making is that the process kills the living insect. At which stage and how does this killing occur?

(A) the adult moths are crushed while laying eggs on mulberry leaves

(B) the larvae are starved by withholding mulberry leaves

(C) the eggs are boiled before they ever hatch into larvae

(D) the pupa inside the cocoon is killed when the cocoon is boiled before the moth can emerge

21. A breeder records that after ten generations of breeding only the finest-fleeced rams and ewes, the flock's average fibre is markedly finer. The mechanism that best explains this gradual change is:

(A) fine-fleece parents tend to pass fine-fleece traits to their offspring, so selecting them each generation shifts the whole flock toward finer wool

(B) the act of shearing the fine sheep forces the remaining sheep to grow finer wool

(C) the climate slowly converted the coarse wool protein into a finer one

(D) the lambs learned to grow finer wool by copying the finest sheep

22. In the wool sequence, exactly one step takes loose cleaned, sorted fibres and twists them into a continuous strand strong enough to weave. That step is:

(A) scouring
(B) shearing
(C) sorting
(D) spinning

23. Cotton and wool reach the spinner as many short fibres that must be twisted together, but reeled silk comes off the cocoon as one very long unbroken strand. This difference means silk is best described as a:

(A) short staple fibre, just like cotton

(B) man-made fibre that is extruded only in short lengths

(C) fibre built from many tiny scales locked end to end

(D) continuous filament fibre, which is why it can be reeled off as a single long thread

24. Besides sheep, several animals yield wool. Which group lists ONLY wool-yielding animals (and no silk producer)?

(A) sheep, mulberry silkworm, goat
(B) sheep, goat, yak, camel
(C) silk moth, sheep, tasar worm
(D) goat, eri silkworm, sheep

25. Order this full silk journey correctly: weaving, cocoon, reeling, mulberry-leaf feeding. The correct sequence from start to fabric is:

(A) cocoon → mulberry-leaf feeding → weaving → reeling

(B) reeling → weaving → cocoon → mulberry-leaf feeding

(C) mulberry-leaf feeding → cocoon → reeling → weaving

(D) weaving → reeling → mulberry-leaf feeding → cocoon

26. A fleece is examined and found to contain coarse outer hairs and fine inner hairs. For making the softest cloth, the processing must:

(A) dye the coarse and fine hairs the same colour so they feel alike

(B) scour the fleece, which automatically removes all coarse hairs

(C) shear the sheep again to grow only fine hairs next time

(D) sort the fleece so the fine inner (undercoat) hairs are separated from the coarse outer guard hairs

27. A claim states: 'Mulberry leaves are fed to sheep to make their wool silkier.' The correct judgement is that the claim is:

(A) false — mulberry leaves are the food of the silkworm (larva), not of sheep, and diet does not turn wool into silk

(B) true — mulberry leaves transfer silk protein into the sheep's wool

(C) true — all fibre animals eat mulberry to improve their fibre

(D) partly true — only Merino sheep benefit from mulberry leaves

28. Comparing the warmth of fabrics, which statement is the most scientifically accurate?

(A) wool warms you by generating heat through a chemical reaction in the fibre

(B) wool is warm because its dark dye absorbs heat from the body

(C) wool is warmer than other fibres because it weighs more

(D) wool keeps you warm by trapping air; it does not produce heat itself, it only slows the loss of body heat

29. Identify the WRONG statement about the silk moth's life cycle.

(A) The larva feeds on mulberry leaves and grows rapidly

(B) The cocoon is spun by the larva as it changes into a pupa

(C) The adult moth spins the silk cocoon around itself before it can fly

(D) The egg hatches into a larva, which is the silkworm

30. A weaver receives reeled silk thread and must turn it into a saree. The next step is weaving, in which:

(A) the cocoon is boiled again to release more thread

(B) the thread is twisted around the silkworm to re-form a cocoon

(C) lengthwise (warp) and crosswise (weft) threads are interlaced on a loom

(D) the silk is washed in cool detergent like a wool sweater

31. Two fabrics are tested: one shrinks and mats in hot water and smells of burnt hair when flamed; the other tears wet and burns with a papery smell. The first fabric is most likely _____ and the second _____.

(A) cotton; wool, because cotton smells of burnt hair

(B) silk; nylon, because both are synthetic

(C) nylon; wool, because nylon shrinks in hot water

(D) wool (a protein fibre); cotton (a plant cellulose fibre)

32. Why does scouring come BEFORE dyeing in wool processing, and not after?

(A) the grease and dirt must be washed out first, or the dye cannot soak evenly into the fibre

(B) dye is what loosens the grease, so it must be applied first

(C) scouring adds the colour, so dyeing afterward would be pointless

(D) the order is arbitrary because both steps clean the wool

33. A pure-silk thread and a pure-wool thread are compared chemically. The correct statement is:

(A) both are proteins, but silk is fibroin and wool is keratin — different proteins

(B) both are cellulose, the same as cotton(C) silk is a protein but wool is cellulose

(D) both are the same protein, so they behave identically

34. Yak wool, camel wool and Pashmina (Angora goat) all share which key feature relevant to where they are reared?

(A) they are all spun by larvae that feed on mulberry leaves

(B) they are all plant fibres harvested from cold-climate crops

(C) they are all synthetic fibres designed to mimic warmth

(D) they come from animals adapted to cold climates, whose thick coats provide warm fibre

35. A sericulturist notices the cocoons are starting to crack as moths chew their way out. To save the silk, the BEST immediate action is to:

(A) let all the moths fly out, since the empty cocoon is the silk

(B) harvest and treat the cocoons with heat/steam before the moths emerge, so the long thread stays unbroken

(C) feed the moths more mulberry leaves to make them stop chewing

(D) boil the cocoons after the moths leave to recover the whole thread

36. A burn test is used to tell a natural ANIMAL fibre from a synthetic like nylon. Which single result most directly marks a sample as an animal fibre rather than nylon?

(A) it melts and draws into a hard, plastic-like bead

(B) it does not catch fire at all (C) it dissolves instantly in cold water

(D) it chars and smells of burnt hair, leaving a crushable ash rather than a hard melted bead

37. In a wool mill, which single step carries the HIGHEST risk of sorter's disease, and why?

(A) rolling the finished, dyed yarn, because dye fumes carry the spores

(B) wearing the finished sweater, because body heat activates the spores

(C) sorting raw, unwashed fleece, because the worker handles dusty fleece that may carry anthrax spores

(D) buying wool from a shop, because shop air is full of spores

38. A silk farmer deliberately lets the moths emerge from a few selected cocoons instead of reeling those cocoons. The most sensible reason is that:

(A) those moths lay the eggs needed to start the next generation of silkworms

(B) the moths themselves are spun into extra silk thread

(C) the moths eat the leftover mulberry leaves so none is wasted

(D) the moths' wings are the source of the very finest silk

39. A care label says 'machine wash warm' for an 80% polyester / 20% cotton shirt but 'cool wash, do not wring' for a 100% wool jumper. The best reason the two labels differ is:

(A) the cotton-polyester shirt contains protein scales that mat, so it needs the gentler care

(B) wool's scaly protein fibres felt and shrink with heat and agitation, while the polyester-cotton blend resists felting, so it tolerates warm machine washing

(C) polyester is an animal fibre and so must always be washed cool

(D) the labels are swapped — wool actually tolerates hot water better than synthetics do

40. Shearing, the FIRST step of wool processing, involves:

(A) washing the grease and dirt out of the fleece

(B) separating the fine inner hairs from the coarse guard hairs

(C) twisting the cleaned fibres into a yarn

(D) cutting off the sheep's fleece — the hair together with a thin layer of skin — usually without hurting the animal

41. Muga silk's natural golden colour and its strong link to Assam best illustrate that:

(A) all silk is naturally golden until it is bleached white

(B) the golden colour comes from boiling the cocoon in dye

(C) different regions and silkworm types produce distinct natural silks, with Assam famous for golden muga

(D) muga is a wool, coloured by the goat's diet in Assam

42. Which classification is correct for the fibres cotton, wool, silk and jute?

(A) cotton and wool are animal fibres; silk and jute are plant fibres

(B) all four are animal fibres because all come from living things

(C) silk and jute are animal fibres; cotton and wool are plant fibres

(D) wool and silk are animal fibres; cotton and jute are plant fibres

43. A worker in a wool unit develops a serious lung illness traced to inhaled spores. The disease and its bacterial cause are:

(A) the common cold, caused by a virus in the dye

(B) sorter's disease (anthrax), caused by Bacillus anthracis spores in raw fleece

(C) malaria, caused by a parasite in the scouring water

(D) tooth decay, caused by bacteria in the wool fibres

44. Sericin and fibroin are both found in raw silk. Which statement correctly distinguishes them?

(A) fibroin is the silk filament that becomes the fabric; sericin is the gum coating that is removed during reeling

(B) sericin is the silk thread woven into cloth; fibroin is the gum thrown away

(C) both are gums removed before reeling, leaving no thread behind

(D) fibroin is a plant cellulose and sericin is an animal protein

45. A sheep is sheared in early summer. By the time the next winter arrives, the same sheep has:

(A) grown a fresh fleece again, which is why a sheep can be sheared year after year

(B) lost the ability to grow wool and become useless for fleece

(C) begun to grow silk in place of wool

(D) developed a skin that turns into cotton fibre

46. If reeling were attempted on a cocoon WITHOUT first softening it in hot water, the most likely problem would be that:

(A) the thread would unwind too fast and tangle into yarn by itself

(B) the silk would lose its colour because hot water sets the dye

(C) the sericin gum would still bind the filament tightly, so the thread would break instead of unwinding smoothly

(D) the pupa would survive and spin a second cocoon

47. A label claims a sweater is '100% Pashmina.' Tracing the fibre to its source, it came from:

(A) the fine undercoat of a Pashmina/Angora goat

(B) the cocoon spun by a mulberry silkworm(C) the seed hairs of a cotton plant

(D) the boiled gum recovered from silk cocoons

48. A poster lists hazards. Which correctly links a wool/silk job to its specific occupational risk?

(A) wool sorters risk anthrax (sorter's disease) from inhaling spores in raw fleece

(B) silk reelers risk anthrax from breathing dye fumes

(C) sweater shoppers risk anthrax from wearing finished wool

(D) shepherds risk anthrax from drinking the sheep's milk only

49. Which reasoning chain about a sheep's fleece is fully correct?

(A) a hot climate makes sheep grow thick fleece → to release more heat → cooling the sheep

(B) a cold climate makes sheep grow thick fleece → the fleece traps air → the trapped air slows heat loss → the sheep stays warm

(C) the fleece produces its own heat → which warms the air → so the sheep needs no body heat

(D) the fleece is plant cellulose → it absorbs sunlight → and burns it for warmth

50. Why does sorting necessarily come BEFORE spinning fine yarn, but AFTER scouring?

(A) sorting cleans the grease, so it replaces scouring entirely

(B) sorting adds the dye, so it must follow spinning

(C) you sort cleaned wool (so scouring comes first), and you must separate fine from coarse hairs before spinning a fine yarn

(D) sorting twists the fibres into yarn, so spinning is unnecessary

51. Making a single silk saree uses the thread of many thousands of cocoons. This fact best explains why:

(A) silk is normally cheaper than cotton

(B) silk sarees are really wool blended with cocoons

(C) one cocoon alone can supply several sarees

(D) silk is relatively expensive and prized, since each garment uses the output of a great many silkworms

52. A student lists wool steps as: shearing → scouring → dyeing → sorting → picking burrs → spinning. Identify the error.

(A) shearing should come last, after the wool is spun

(B) scouring should come before shearing

(C) dyeing is placed too early — it should come after sorting and picking burrs, not before them

(D) spinning should come first, before shearing

53. A waste-sorting project separates fibres that rot away in soil from those that do not. Into the 'does NOT rot / not biodegradable' bin should go:

(A) cotton and wool

(B) nylon and polyester

(C) silk and cotton

(D) wool and jute

54. A documentary shows cocoons being dropped into boiling water while the larva-spun thread is unwound onto a reel. The two things the boiling water is achieving at once are:

(A) softening the sericin so the thread unwinds, and killing the pupa so the single thread is not broken

(B) dyeing the silk gold and hatching the eggs faster

(C) feeding the pupa and washing dust off the fabric

(D) turning the silk into wool and adding the final twist

55. A trader sells 'animal-fibre yarn' that turns out to weaken sharply when wet, burns with a papery smell, and never shrinks in hot water. The honest conclusion is:

(A) it is genuine wool, since wool weakens when wet

(B) the yarn is mislabelled — it behaves like cotton, a plant fibre, not an animal fibre

(C) it is genuine silk, since silk burns with a papery smell

(D) it is genuine Pashmina, since Pashmina never shrinks

56. Which statement about where India's wool and silk come from is correct?

(A) wool sheep are reared mainly in the hot coastal plains, and muga silk comes from Punjab

(B) wool sheep are reared in hilly states like Jammu & Kashmir and Himachal Pradesh, while muga silk is famous in Assam

(C) both wool and silk come only from the Himalayan glaciers

(D) wool comes from Assam's silkworms and silk from Kashmir's sheep

57. A sericulture unit feeds its silkworms only wilted, dried leaves and few in number. The most likely consequence for the silk is that:

(A) the larvae spin more silk to make up for the poor food

(B) the silk turns into wool because the diet changed

(C) poorly fed larvae spin smaller, poorer cocoons, yielding less and weaker silk

(D) the adult moths spin extra cocoons to compensate

58. Why is a finished, scoured and dyed woollen sweater essentially safe from sorter's disease, even though the disease comes from sheep wool?

(A) because the disease can only infect sheep, never humans

(B) scouring and full processing remove the anthrax spores from the fleece before it ever reaches the sweater

(C) because the dye on the sweater is an antibiotic that cures anthrax

(D) because finished sweaters are actually made of cotton, not wool

59. Order the silk-moth life cycle and identify which arrow marks the silk-spinning event: egg → larva → pupa → adult moth.

(A) silk is spun at the egg → larva transition, as the egg hatches

(B) silk is spun at the pupa → adult moth transition, as the moth emerges

(C) silk is spun by the adult moth after it lays its eggs

(D) silk is spun at the larva → pupa transition, when the larva spins its cocoon

60. A factory manager wants to maximise BOTH wool quality and worker safety with the fewest changes. Which combined plan is soundest?

(A) use a coarse breed, skip sorting, and let sorters work without masks to save money

(B) skip scouring to keep the grease, and dye the wool brighter to hide any spores

(C) rear a fine-wool breed, sort the fleece carefully, and protect sorters with disinfected fleece, masks and ventilation

(D) feed the sheep mulberry leaves and boil the fleece to kill any pupae

Solutions — Science (Class 7), Chapter 3: Fibre to Fabric — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	A	31	D	41	C	51	D
2	C	12	D	22	D	32	A	42	D	52	C
3	C	13	B	23	D	33	A	43	B	53	B
4	B	14	D	24	B	34	D	44	A	54	A
5	C	15	B	25	C	35	B	45	A	55	B
6	A	16	A	26	D	36	D	46	C	56	B
7	A	17	B	27	A	37	C	47	A	57	C
8	B	18	B	28	D	38	A	48	A	58	B
9	C	19	D	29	C	39	B	49	B	59	D
10	C	20	D	30	C	40	D	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Protein fibres (wool, silk = keratin/fibroin) contain nitrogen and smell of burnt hair when burned; cellulose fibres (cotton, jute) smell like burning paper and leave soft ash — so X is animal, Y is plant. The burnt-hair smell belongs to protein fibres, not cotton, so the first distractor reverses the test. Only X charring like hair is the animal one, so 'both are animal' ignores Y's papery, plant signature. Silk is a protein fibre and would smell of burnt hair like X, not give a papery ash, so calling Y silk misreads the result.

2. (C) Wool is hair that grows as a sheep's body covering and is sheared off the living animal, whereas silk is a fluid the silkworm larva secretes from its body that hardens in air into a long filament — two completely different biological routes to an animal fibre. Wool is the sheep's secreted fluid and silk the silkworm's body hair reverses both origins. Neither is spun by leaf-eating insects in the wool's case — only the silkworm eats mulberry, and the sheep does not spin. Both are animal fibres, so 'wool from a plant' is simply wrong.

3. (C) Wool warms by trapping a layer of still air; the loose, crimped blanket P holds more air, and air conducts heat poorly, so it insulates better. The two blankets are equal in mass, so 'heavier' is false and weight is not what insulates anyway. Compressing wool into felt does not change its protein into cotton — it stays keratin. Wool does not produce heat; it only slows the loss of the body's heat, so 'generates its own heat' is wrong.

4. (B) Sheep grow a thick fleece as protection against cold, so cool highlands (J&K, Himachal) are the classic Indian wool-rearing belt. Heat does not make sheep grow more wool 'to stay cool' — fleece is for warmth, so the lowland claim inverts the biology. Climate clearly affects fleece growth, so 'no effect' is false. Wool comes from several animals including sheep (the main source), not only goats.

5. (C) After shearing, the fleece is scoured (washed), then sorted by quality, then burrs are picked out, and only then is the clean fibre dyed — scouring → sorting → picking burrs → dyeing. Dyeing is near the end, not first, so it cannot lead. Scouring must precede sorting because you sort cleaned wool, so 'sorting first' is wrong. Burr-picking and dyeing both come after the wool is washed, so starting with burr-picking before scouring misorders it.

6. (A) Selective breeding chooses parent sheep with desirable fleece so each generation yields finer, softer, more uniform wool. Sheep produce wool, not silk — silk comes from the silkworm — so breeding cannot make a sheep spin silk. Breeding for quality does not eliminate the fleece; the animal still grows wool that must be sheared. Wool's keratin protein cannot be converted to cellulose by breeding, so that option is biologically impossible.

7. (A) Between egg and pupa is the larva (the silkworm/caterpillar), which feeds hungrily on mulberry leaves and grows before spinning a cocoon. The cocoon is the protective case the larva spins, not a life-cycle stage that lays eggs — the adult moth lays eggs. The adult moth is the final stage and does not spin silk; the larva does. The pupa forms after the larva spins its cocoon, so it cannot hatch directly from the egg before the larva.

8. (B) A cocoon is one long unbroken silk thread, often hundreds of metres long, wound by the larva around its body — that is why reeling can unwind a single continuous fibre. It is not a felt of many short random threads; that random arrangement describes wool felting, not a cocoon. It is spun, not sheared like a fleece, so 'outer fleece' confuses silk with wool. It is a silk case the larva makes, not a leftover egg shell.

9. (C) The question asks for the false reason: shears are tools that work in any season, so 'mechanically impossible in winter' is the invalid justification (the correct answer here). It is true that summer shearing lets a fresh fleece regrow before winter. It is true that shearing in winter would dangerously expose the sheep to cold. It is true that in warm months the sheep does not need its fleece for warmth, so heat loss is not a problem.

10. (C) Wool fibres have tiny scales; hot water plus rough agitation makes these scales interlock, so the sweater shrinks and becomes matted. It does not stretch larger — felting tightens, not loosens, the fibres. Heat cannot convert wool's keratin protein into plant cellulose, so it never becomes cotton. Wool is clearly affected by hot water (that is why the label warns against it), so 'nothing happens' is wrong.

11. (A) Scouring first removes grease and dirt so the burrs become visible and easy to pick; doing it on greasy fleece hides the burrs and many are missed. Grease does not 'lubricate' burrs out — it sticks dirt and plant bits in place. Step order matters in processing, so 'no difference' is false. Burr-picking has nothing to do with dyeing, which is a separate later step, so no automatic dyeing occurs.

12. (D) Scouring (a wool step) washes out grease/dirt; reeling (a silk step) uses hot water to soften the sericin gum so the cocoon's single thread can be unwound. Neither step is for dyeing — colour is added in a separate dyeing step. Scouring belongs to wool and reeling to silk, so swapping them ('scouring unwinds silk') is wrong. Adding the twist to make yarn is a different spinning/rolling step, so 'both only add the twist' is false.

13. (B) Tasar, eri and muga are India's well-known non-mulberry (wild) silks, spun by silkworms other than the mulberry one. They are natural animal fibres from silkworms, not petroleum-based synthetics like nylon/polyester. They are silks, not wool grades, so they have nothing to do with sorting fleece. They are types of silk, not life-cycle stages (which are egg, larva, pupa, moth).

14. (D) The larva spins silk to build a protective case — the cocoon — around itself while it transforms into the soft, defenceless pupa, sheltering it during that vulnerable stage. The cocoon is not a food store; the adult moth that emerges does not feed at all. It is not for attracting a mate — mating belongs to the adult stage, not the larva. It does not trap leaves for the pupa, because the pupa does not feed; the cocoon's role is purely protection.

15. (B) An emerging moth breaks the cocoon, snapping the single long filament into short fragments; killing the pupa with heat keeps the thread whole for reeling. Heat does not make the pupa grow into a bigger silk-spinning moth — the larva already spun the silk before pupating. Cleaning is incidental, not the main purpose; preserving the unbroken thread is. Boiling softens the gum but does not dye the silk gold; muga's gold colour is natural, not from boiling.

16. (A) One cocoon's filament is extremely fine and weak, so several are reeled together to make a thread strong and thick enough to use. A single filament is too thin, not too thick — the opposite of the first distractor. Combining is for strength, not colour-mixing; mulberry silk is already white, so 'mixing for white' is false. A cocoon's thread is silk, not wool, so there is no wool to blend in.

17. (B) Sorter's disease comes from inhaling anthrax spores in raw fleece dust; the sorter is exposed to raw fleece, while a finished sweater has been scoured and processed, removing the spores. The risk is from handling/inhaling raw fleece, not from wearing finished wool, so 'only attacks wearers' is backwards. Wool sweaters are made of wool, not cotton, so that claim is simply false. Dye does not cure anthrax; safe processing, not dye, is what protects the shopper.

18. (B) Anthrax spores are chiefly inhaled in the dust of raw, unwashed fleece, so a dust mask blocks that main entry route; gloves alone protect the hands but leave the breathing passages — the real route — unguarded. The disease is not skin-cuts-only, so 'gloves alone are enough' is false. It is not caught from drinking sheep's milk, so 'neither helps' is wrong. The main route is inhalation, not the eyes, so goggles-only misidentifies the route.

19. (D) The Angora/Pashmina goat yields fine wool and the silk moth (its larva) yields silk — both correct. Sheep give wool and the silk moth gives silk, so 'sheep → silk; silk moth → wool' swaps them. The mulberry plant only feeds the silkworm — silk comes from the worm, not the plant — and sheep give wool, not cotton, so that pairing is doubly wrong. The yak gives wool, not silk, and the camel gives wool, not cotton, so the last pairing is wrong on both counts.

20. (D) Cocoons are boiled while the pupa is still inside, killing it so the long thread stays unbroken — that is the ethical objection. Adult moths are not crushed during egg-laying; the killing happens at the cocoon/pupa stage. Larvae are fed mulberry leaves, not starved — feeding is essential to making them spin good cocoons. Eggs are not boiled; the eggs must hatch into the silk-spinning larvae first, so the killing targets the pupa, not the egg.

21. (A) Fleece quality is inherited, so sheep with fine fleece tend to have fine-fleeced offspring; choosing such animals as parents each generation gradually shifts the whole flock toward finer wool — that is selective breeding. Shearing a sheep does not change the wool grown by other sheep, so the second option has no mechanism. Climate does not transmute one wool protein into a finer one within a flock. Lambs cannot 'learn' to grow finer wool by copying, since fibre fineness is inherited, not learned.

22. (D) Spinning is the step that twists the loose, cleaned and sorted fibres together into a continuous strand (yarn) strong enough to be woven — that is exactly its job. Scouring only washes out grease and dirt; it does not twist fibres into yarn. Shearing merely cuts the fleece off the sheep, long before any yarn is made. Sorting separates fine from coarse hairs but leaves them as loose fibre, not a twisted strand.

23. (D) A cocoon is wound from one extremely long, unbroken strand, so silk is a continuous filament fibre — that is precisely why reeling can draw it off as a single long thread. Cotton's short staple fibres are the opposite of one long filament, so 'short staple like cotton' is wrong. Silk is natural, not a man-made extruded fibre, and it is long, not short. The tiny overlapping scales belong to wool's structure, not to a smooth silk filament, so 'scales locked end to end' misdescribes silk.

24. (B) Sheep, goat, yak and camel are all wool-yielding animals, so that group is entirely wool with no silk producer. The mulberry silkworm gives silk, not wool, so the group containing it is wrong. The silk moth and the tasar worm are silk producers, so that group mixes in silk. The eri silkworm likewise gives (eri) silk, not wool, so its group is wrong too.

25. (C) The larva first feeds on mulberry leaves, then spins a cocoon; the cocoon is reeled into thread, which is finally woven into fabric — feeding → cocoon → reeling → weaving. Feeding must come before the cocoon (the fed larva spins it), so any order placing the cocoon first is wrong. Reeling and weaving are the last two steps, so beginning with them inverts the whole sequence. Weaving is the final step, so it cannot start the chain.

26. (D) A fleece naturally mixes coarse guard hairs and fine undercoat; sorting separates the fine hairs needed for soft cloth. Dyeing only colours fibres and cannot change a coarse hair's feel, so equal colour does not make them feel alike. Scouring washes out grease and dirt but does not pick out coarse hairs — that is sorting's job. Re-shearing the same sheep regrows the same mix of hairs, so it does not yield only fine hairs.

27. (A) Mulberry leaves feed the silkworm larva so it can spin silk; sheep do not eat them to improve wool, and no diet converts wool into silk. There is no 'silk protein' that passes from leaf to wool — wool is keratin made by the sheep regardless of mulberry. Not all fibre animals eat mulberry; it is specifically the silkworm's food. No sheep breed, Merino included, gains silkiness from mulberry, so the 'partly true' option is also false.

28. (D) Wool insulates by trapping a layer of still air around the body, slowing heat loss — it never makes its own heat. There is no heat-generating chemical reaction in wool fibres, so that claim is false. Warmth comes from trapped air, not from dye colour, and natural fleece is creamy, not necessarily dark. Weight is not what insulates — a light, lofty wool can trap more air than a heavier, denser fabric.

29. (C) The silk is spun by the larva, not the adult moth — the moth has already passed the silk-spinning stage — so this is the false statement (the correct answer). It is true that the larva feeds heavily on mulberry leaves and grows. It is true that the larva spins the cocoon as it transforms into a pupa. It is true that the egg hatches into the larva, which is the silkworm.

30. (C) Weaving interlaces lengthwise warp threads with crosswise weft threads on a loom to make cloth. Boiling the cocoon belongs to reeling, which is already done before weaving, so it is not the weaving step. Thread is not wound back onto a silkworm to re-form a cocoon — that reverses the process nonsensically. Cool-water detergent washing is wool-care advice, not the weaving of silk into a saree.

31. (D) Shrinking/matting in hot water plus a burnt-hair smell are wool's signatures; weakening when wet and a papery burn smell are cotton's — so wool then cotton. Cotton does not smell of burnt hair (it smells like burning paper), so the first distractor swaps the two. Silk is natural and nylon synthetic, but neither shrink-and-mat behaviour nor a burnt-hair smell points to nylon, so that pairing fails. Nylon is a synthetic, not the wool-like fibre that mats and smells of hair, so 'nylon; wool' misassigns the first fabric.

32. (A) Clean fibre takes dye evenly, so scouring (washing out grease/dirt) must precede dyeing. Dye does not loosen grease — soap and water in scouring do that — so applying dye first would just sit on a greasy surface. Scouring cleans; it does not add colour, so 'scouring adds the colour' is false. The order is not arbitrary, and only scouring cleans while dyeing colours, so they are not interchangeable.

33. (A) Wool and silk are both proteins but chemically distinct — wool is keratin, silk is fibroin. Neither is cellulose; cellulose is the plant-fibre material in cotton, so 'both cellulose' is wrong. Wool is a protein (keratin), not cellulose, so calling wool cellulose mis-classes it. They are different proteins arranged differently, which is why they behave differently — so 'same protein, identical' is false.

34. (D) Yak, camel and Pashmina goat are cold-climate animals whose thick coats give warm wool, which is why such wool comes from highland/cold regions. None are spun by mulberry-eating larvae — that describes silkworms, not these wool animals. They are animal fibres, not plant fibres from crops. They are natural, not synthetic, so 'designed to mimic warmth' is wrong.

35. (B) An emerging moth snaps the single thread, so cocoons must be treated with heat before emergence to keep the filament whole. The empty, chewed cocoon is not intact silk — the thread is already cut into pieces, so letting moths out ruins it. Adult moths do not eat mulberry leaves (the larva does), so feeding them does nothing. Boiling after the moths leave is too late — the thread is already broken, so the whole continuous thread cannot be recovered.

36. (D) Protein (animal) fibres char and give a burnt-hair smell, leaving a soft crushable ash; a synthetic like nylon instead melts and pulls into a hard plastic-like bead — so the burnt-hair-and-ash result marks the animal fibre. Melting into a hard bead is the nylon signature, the opposite of an animal fibre. 'Does not burn at all' fits neither and is not a test result here. Animal fibres do not dissolve instantly in cold water, so that result identifies nothing.

37. (C) Sorting handles raw, unwashed fleece whose dust can carry anthrax spores, so that step is the riskiest. Rolling finished yarn happens after scouring has removed spores, and dye fumes do not carry anthrax, so that step is low risk. Wearing a finished sweater is safe because processing removed the spores; body heat does not 'activate' them. Buying from a shop involves only processed wool, so shop air is not a spore source.

38. (A) A few cocoons are spared so their moths can emerge, mate and lay the eggs that start the next generation of silkworms — without them, rearing could not continue. The moths are not spun into silk; the silk is the larva's cocoon thread, already made. The reason is breeding, not tidying up leftover leaves. The moths' wings are not a silk source — silk comes only from the larva's cocoon, so that option is false.

39. (B) Wool's fibres carry tiny scales that interlock under heat and agitation, so wool felts and shrinks and needs cool, gentle washing; a polyester-cotton blend has no such scaly protein and resists felting, so it safely takes a warm machine wash — hence the different labels. The shirt has no protein scales to mat, so the first option mis-sources the risk. Polyester is a synthetic, not an animal fibre, so 'must be washed cool' for that reason is wrong. The labels are not swapped — wool tolerates hot water far less well than synthetics, so the last option inverts the truth.

40. (D) Shearing is the removal of the fleece — the sheep's hair together with a thin layer of skin — cut off with clippers, normally without hurting the animal; it is the first step before any cleaning or processing. Washing grease out is scouring, a later step, not shearing. Separating fine from coarse hairs is sorting, also later. Twisting cleaned fibres into yarn is spinning, the final step — none of these is what shearing means.

41. (C) Muga is a non-mulberry silk with a natural golden colour, regionally tied to Assam — showing silk type and region shape the fibre. Most silk (mulberry) is naturally white, not golden, so 'all silk is golden until bleached' is false. Muga's gold is natural, not added by boiling in dye. Muga is a silk from a silkworm, not a wool from a goat, so the diet explanation is wrong.

42. (D) Wool (sheep) and silk (silkworm) are animal fibres; cotton (seed hairs) and jute (plant stem) are plant fibres — the correct split. Cotton is a plant fibre, not animal, so grouping it with wool as animal is wrong. Plants are living too, but cotton and jute are plant fibres, so 'all four animal' is false. Silk is animal and jute is plant, so pairing silk with jute as animal misclassifies both jute and cotton.

43. (B) Sorter's disease is anthrax, caused by Bacillus anthracis spores carried in raw fleece and inhaled by sorters. The common cold is a mild viral illness unrelated to wool dust, and dye carries no cold virus. Malaria is mosquito-borne by a parasite, not from scouring water. Tooth decay involves mouth bacteria on teeth, having nothing to do with inhaled fleece spores.

44. (A) Fibroin is the actual silk filament that becomes cloth; sericin is the sticky gum coating that is softened and removed in reeling. The roles are not reversed — sericin is the discarded gum, not the woven thread, so the first distractor swaps them. Not both are removed; the fibroin filament remains as the thread, so 'no thread behind' is false. Both are animal proteins of silk; neither is plant cellulose, so the last option mis-classes fibroin.

45. (A) Wool is hair that keeps growing, so once a sheep is sheared in summer it grows a fresh fleece again before winter — which is why the same sheep can be sheared season after season, making wool a renewable fibre. A sheared sheep does not lose the ability to grow wool, so 'useless for fleece' is false. Shearing does not switch the sheep to producing silk. Sheep skin does not turn into cotton fibre, which is a plant product, so the last option is biologically impossible.

46. (C) Hot water softens the sericin gum so the single filament can unwind; without it the gum holds the thread fast and it breaks rather than reeling off. It would not unwind faster — the gum makes unwinding harder, not easier. Hot water in reeling softens gum, it does not set dye, so colour loss is not the issue. The hot-water step also kills the pupa; skipping it would not let a pupa spin a second cocoon — a pupa does not spin silk anyway.

47. (A) Pashmina is the fine soft undercoat of the Pashmina (Angora) goat — a wool, an animal fibre from a goat. A silkworm cocoon yields silk, not Pashmina wool, so that source is wrong. Cotton seed hairs are a plant fibre, unrelated to a goat's wool. Boiled cocoon gum is sericin, a silk by-product, not goat wool.

48. (A) Sorter's disease (anthrax) specifically threatens those who handle raw fleece — the sorters — by inhaling its spores. Silk reelers work with boiled cocoons, not raw fleece, and dye fumes do not carry anthrax, so that link is false. Finished sweaters are processed and spore-free, so shoppers are not at risk. The recognised occupational route is inhaling raw-fleece spores, not 'drinking milk only,' so that option misstates the risk.

49. (B) Cold prompts a thick fleece, which traps air, and trapped air (a poor conductor) slows the sheep's heat loss, keeping it warm — a sound chain. Thick fleece is for warmth in cold, not for releasing heat in hot climates, so the first chain inverts the purpose. Fleece traps the body's heat; it does not generate its own heat, so the second chain is false. Wool is animal keratin, not plant cellulose, and it does not 'burn sunlight,' so the last chain is wrong throughout.

50. (C) Sorting separates fibres by quality and is done on already-cleaned wool, so scouring precedes it; and the fine hairs must be picked out before they can be spun into fine yarn. Sorting does not clean grease — scouring does — so it cannot replace scouring. Sorting does not dye wool; dyeing is a separate later step, so 'sorting adds dye' is wrong. Sorting separates fibres; it does not twist them into yarn, so spinning is still needed.

51. (D) Because each garment uses the silk of many thousands of cocoons — and each cocoon is one silkworm's whole effort — silk is comparatively costly and prized; the sheer number of insects behind a single saree explains its value. Silk is dearer, not cheaper, than cotton, so the first option is wrong. A silk saree is silk, not wool blended with cocoons. One cocoon yields only a little thread — nowhere near several sarees — so that option inverts the scale.

52. (C) The fibre must be sorted and have burrs removed while still natural-coloured, then dyed; placing dyeing before sorting/burr-picking is the error. Shearing is the first step (removing the fleece), not the last, so 'shearing last' is wrong. You cannot scour a fleece that is still on the sheep, so scouring cannot precede shearing. Spinning is the final step that makes yarn, so it cannot come before shearing.

53. (B) Nylon and polyester are man-made synthetics that do not rot in soil, so they belong in the 'does not biodegrade' bin. Cotton and wool are natural fibres that rot away, so 'cotton and wool' is wrong. Silk and cotton are both natural and biodegradable, so that pair does not belong in the non-rotting bin. Wool and jute are likewise natural fibres that decompose, so the last pair is wrong too.

54. (A) Boiling softens the sericin gum (so the filament reels off) and kills the pupa (so it cannot bite through and break the thread) — both at once. It does not dye silk gold or hatch eggs; muga's gold is natural and eggs are not in the cocoon. It does not feed the pupa (it kills it) or wash fabric (there is no fabric yet, only thread). It cannot turn silk into wool, and the twist is added later in spinning/weaving, not by boiling.

55. (B) Weakening when wet, a papery burn smell, and no shrink-felting are all cotton (plant) traits, so an 'animal-fibre' label is false. Wool actually shrinks/mats in hot water and smells of burnt hair, so these traits are not wool's. Silk is a protein fibre and burns with a burnt-hair (not papery) smell, so the papery burn rules out silk. Pashmina is wool and would mat in hot water; 'never shrinks' plus a papery burn does not fit Pashmina either.

56. (B) Wool sheep thrive in cool hill states (J&K, Himachal), and muga silk is the pride of Assam — both correct. Wool sheep are reared in cool hills, not hot coastal plains, and muga is from Assam, not Punjab, so that option is wrong on both counts. Neither fibre comes 'only from glaciers' — they come from animals reared in various regions. Wool comes from sheep and silk from silkworms; swapping them (Assam's silkworms giving wool, Kashmir's sheep giving silk) is backwards.

57. (C) Silkworm larvae must feed well (on fresh mulberry) to grow and spin good cocoons; poor feeding gives smaller, poorer cocoons and less, weaker silk. Hunger weakens larvae rather than making them spin more, so 'more silk to compensate' is false. Diet cannot turn silk into wool — silk is fibroin from the silkworm regardless of feeding. Adult moths do not spin cocoons (the larva does), so they cannot 'compensate' by spinning.

58. (B) Processing — especially scouring — washes the anthrax spores out of the fleece, so a finished sweater carries essentially none, making it safe. Anthrax can infect humans (that is the whole point of sorter's disease), so 'only sheep' is false. Dye is a colourant, not an antibiotic, so it does not cure anthrax. Woollen sweaters are made of wool, not cotton, so that explanation is simply incorrect.

59. (D) The fully grown larva spins its silk cocoon as it changes into a pupa, so silk-spinning marks the larva → pupa arrow. At the egg → larva transition the egg merely hatches into a feeding caterpillar; no silk is spun yet. At pupa → moth the cocoon is already made and the moth emerges (it does not spin then). The adult moth lays eggs but never spins silk, so the last option is wrong.

60. (C) Fine-wool breed + careful sorting gives quality, while disinfecting fleece and giving masks/ventilation protects sorters from anthrax — a sound combined plan. A coarse breed with no sorting hurts quality, and no masks endangers sorters, so that plan fails both goals. Skipping scouring leaves greasy, dirty wool and does not remove spores; bright dye cannot hide or kill spores. Mulberry leaves are silkworm food, not sheep food, and a fleece has no pupae to boil, so that plan is irrelevant to wool.